

Lecture 33 — Infinity of $\mathbb{N}$, $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$, and Cantor's Theorem

CS 173: Discrete Structures · Cardinality & the Diagonal Argument

University of Illinois at Urbana-Champaign

1 Setup and Goal

We continue developing the theory of cardinality. Two large claims sit on the horizon:

$$|\mathbb{N}| = |\mathbb{N}_d| = |\mathbb{Z}|, \quad |\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|.$$

Before we can prove either, we need a foundational fact: the set $\mathbb{N}$ itself is infinite. To formalise this we recall that, throughout this course, a natural number $n \in \mathbb{N}$ is treated as the finite set $n = \{0, 1, \dots, n-1\}$, so a statement like $|n| = |\mathbb{N}|$ is a statement about cardinalities of sets.

Our first target is therefore the following.

Remark. We want to show

$$(\forall n \in \mathbb{N})(|n| \neq |\mathbb{N}|),$$

i.e. no finite natural number has the same cardinality as $\mathbb{N}$. Equivalently, $\mathbb{N}$ is infinite.

2 $\mathbb{N}$ is infinite

Theorem 2.1 ($\mathbb{N}$ is infinite). $\mathbb{N}$ is infinite. Equivalently,

$$(\forall n \in \mathbb{N})(|n| \neq |\mathbb{N}|).$$

Proof. Towards a contradiction, suppose there exists $n \in \mathbb{N}$ such that $|n| = |\mathbb{N}|$. Then there exists a bijection

$$f : n \longrightarrow \mathbb{N}.$$

(Recall $n = \{0, 1, \dots, n-1\}$.)

Construct a witness that f misses. Define

$$S := \sum_{i=0}^{n-1} f(i),$$

and observe that $S \in \mathbb{N}$ because each $f(i) \in \mathbb{N}$ and $\mathbb{N}$ is closed under finite sums.

Bound every output of f by S . For any $i \in \{z \in \mathbb{N} \mid z < n\}$, we have

$$f(i) \leq f(i) + \sum_{\substack{j=0 \\ j \neq i}}^{n-1} f(j) = \sum_{j=0}^{n-1} f(j) = S.$$

The first inequality holds because every term $f(j)$ being summed lies in $\mathbb{N}$, hence is non-negative. Therefore

$$(\forall i \in n)(f(i) \leq S < S+1).$$

Derive the contradiction. Since f is surjective and $S + 1 \in \mathbb{N}$, there exists $x \in \{z \in \mathbb{N} \mid z < n\}$ such that

$$f(x) = S + 1.$$

However, by the bound just established, $f(x) < S + 1$, so $f(x) \neq S + 1$. Contradiction.

Therefore no such bijection exists, and $\mathbb{N}$ is infinite. $\square$

Remark (Why this proof works). The argument is a clean cardinality version of the diagonal style: assuming a bijection from a finite set onto $\mathbb{N}$ lets us *construct* a specific natural number, namely $S + 1$, that is strictly larger than every value the bijection produces. Surjectivity then forces a contradiction.

3 $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$

Having established that $\mathbb{N}$ is genuinely infinite, we now compare it against the seemingly larger set $\mathbb{N} \times \mathbb{N}$ of ordered pairs of naturals. Despite appearances, the two have the same cardinality.

Theorem 3.1. $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$.

Proof (in progress). By the Schröder–Bernstein style strategy, it suffices to exhibit two injections, one in each direction.

An injection $f : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Define

$$f(n) := (0, n).$$

This is plainly injective: if $f(n_1) = f(n_2)$, then $(0, n_1) = (0, n_2)$, so $n_1 = n_2$.

An injection $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Define

$$g(x, y) := 2^x \cdot 3^y.$$

We claim g is injective. Suppose $g(x_1, y_1) = g(x_2, y_2)$, i.e.

$$2^{x_1} \cdot 3^{y_1} = 2^{x_2} \cdot 3^{y_2}.$$

By the Fundamental Theorem of Arithmetic, the prime factorisation of a positive integer is unique. Since 2 and 3 are distinct primes, the exponents on each side must agree, giving $x_1 = x_2$ and $y_1 = y_2$. Hence $(x_1, y_1) = (x_2, y_2)$, so g is injective.

With injections in both directions established, the conclusion $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$ follows from the cardinal comparability theorem to be applied next lecture. $\square$

Remark (Why $2^x \cdot 3^y$?). The choice of two distinct primes is what makes the encoding injective. Any two distinct primes p, q would do — for example $g(x, y) = p^x q^y$ — because unique factorisation guarantees that recovering (x, y) from $g(x, y)$ is unambiguous. Picking 2 and 3 is just the smallest, cleanest choice.

Caution. Avoid the temptation to write $g(x, y) = 2^x + 3^y$ or $g(x, y) = x + y$. Neither is injective: e.g. $1 + 2 = 2 + 1$ but $(1, 2) \neq (2, 1)$. Multiplicativity together with distinct prime bases is exactly what unique factorisation needs.

4 Cantor's Theorem

So far we have stacked $\mathbb{N}, \mathbb{Z}, \mathbb{N} \times \mathbb{N}$ on top of each other and found they all share the same cardinality. A natural question: is there *any* operation that genuinely produces a strictly larger set? Cantor's theorem answers yes — the power set always does.

Theorem 4.1 (Cantor's Theorem). $\forall X (|X| < |P(X)|)$.

Proof (the diagonal argument). Let X be a set. Towards a contradiction, suppose $|X| \geq |P(X)|$. Then there exists a function

$$\varphi : X \longrightarrow P(X)$$

such that φ is surjective. Define the *anti-diagonal set*

$$\Delta := \{x \in X \mid x \notin \varphi(x)\}.$$

Observe that $\Delta \subseteq X$, so $\Delta \in P(X)$.

Since φ is surjective and $\Delta \in P(X)$, there exists $\delta \in X$ such that

$$\varphi(\delta) = \Delta.$$

We now ask whether $\delta \in \Delta$, and split into the two exhaustive cases.

Case 1: $\delta \in \Delta$. By definition of Δ , membership requires $\delta \notin \varphi(\delta)$. Since $\varphi(\delta) = \Delta$, this gives $\delta \notin \Delta$, contradicting the case assumption.

Case 2: $\delta \notin \Delta$. Then $\neg(\delta \notin \varphi(\delta))$ fails to hold for δ , i.e. δ does *not* satisfy the membership condition for Δ . Since the only way to fail $x \notin \varphi(x)$ is to have $x \in \varphi(x)$, we get $\delta \in \varphi(\delta) = \Delta$, contradicting the case assumption.

Both cases yield a contradiction, so no surjection $\varphi : X \rightarrow P(X)$ exists. Combined with the obvious injection $x \mapsto \{x\}$ from X into $P(X)$ (which gives $|X| \leq |P(X)|$), we conclude

$$|X| < |P(X)|. \quad \square$$

Remark (The case split, restated cleanly). The whole proof hinges on a single self-referential question: *is δ a member of the set $\varphi(\delta) = \Delta$?* The definition of Δ makes the two possible answers logically swap:

$$\delta \in \Delta \iff \delta \notin \varphi(\delta) \iff \delta \notin \Delta.$$

That biconditional is the contradiction. Cases 1 and 2 just walk through it in each direction.

Remark (Connection to Russell’s paradox). The same self-referential trick produces Russell’s paradox: take X to be “the set of all sets” and let φ be the identity, so Δ becomes the set of all sets that do not contain themselves. The contradiction here is precisely why no such universal set can exist in ZFC. Cantor’s theorem can be read as a constructive cardinality version of the same observation.

Remark (Why “diagonal” / “anti-diagonalising”?). Imagine listing the elements of X down the rows of an infinite table, and labelling the columns with the same elements. The cell at row x , column y records whether $y \in \varphi(x)$. The set Δ is built by walking *down the diagonal* ($x = y$ in each row) and flipping the answer: x goes into Δ exactly when $x \notin \varphi(x)$. By construction, Δ disagrees with $\varphi(x)$ at the entry x for every row x , so it cannot be any row of the table. The same trick powers Cantor’s proof that $|\mathbb{R}| > |\mathbb{N}|$, and Russell’s paradox.

Remark (Reference). The construction is sometimes called the *anti-diagonalising set* in the literature; nLab catalogues it under “diagonal argument”.

5 Summary

1. $\mathbb{N}$ is infinite: no finite natural number n satisfies $|n| = |\mathbb{N}|$. The proof builds the witness $S + 1$ from the sum $S = \sum_{i=0}^{n-1} f(i)$ and uses surjectivity to force a contradiction.
2. Adding pairs does not enlarge cardinality: $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$. The forward injection $n \mapsto (0, n)$ is trivial; the reverse injection $(x, y) \mapsto 2^x \cdot 3^y$ relies on unique prime factorisation.
3. Cardinality arguments at this level have a recipe: *either* construct an explicit bijection, *or* construct two injections and appeal to the cardinal comparability theorem.
4. **Cantor’s theorem:** $|X| < |P(X)|$ for every set X . The proof opens with the anti-diagonal set $\Delta = \{x \in X \mid x \notin \varphi(x)\}$, which by construction cannot be in the image of any surjection $\varphi : X \rightarrow P(X)$.

Remark (Looking ahead). Next lecture will (i) close out the chain $|\mathbb{N}| = |\mathbb{N}_d| = |\mathbb{Z}|$ and (ii) finish combining today’s two injections into the formal $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$ via the cardinal comparability theorem. With Cantor’s theorem now in hand, the canonical application is $|\mathbb{N}| < |P(\mathbb{N})| = |\mathbb{R}|$ — the existence of cardinalities strictly larger than $|\mathbb{N}|$, and the beginning of the infinite hierarchy of infinities.